For the Hardware Path, this is what we are expecting from you. You can watch the videos to get an idea of the type of projects we have in mind. These are only recommendations/examples to help you understand the level and direction of the project we are expecting from you.

https://drive.google.com/file/d/1pkTxVEv8Yih-ADaJkNeLI_99HCiXZ0ql/view?usp=sharing

https://drive.google.com/file/d/13x4Qqz31seJJNjZ7hJRPngnlIPHEjSNM/view?usp=sharing

And, For the Software Path, this is what we are expecting from you: These are only recommendations/examples to help you understand the level and direction of the project we are expecting from you.

1. Crop Recommendation System

Problem: Recommend a suitable crop based on soil and environmental conditions.

Streamlit:

- Enter soil parameters
- Enter temperature and humidity
- Enter rainfall
- Enter soil pH
- Run the ML model
- Display recommended crop
- Display recommendation confidence
- Show the input conditions used
- Provide information about the recommended crop

Sample Output:

Recommended Crop: Rice
Confidence Score: 89%
Soil pH: 6.4
Rainfall: 210 mm

2. Road Pothole Detection & Severity Analysis

Problem: Detect potholes in road images and classify their severity to assist road maintenance authorities.

Streamlit:

- Upload road image/video frame

- Detect potholes
- Draw bounding boxes around detected potholes
- Display confidence scores
- Estimate/classify pothole severity
- Display total number of potholes
- Provide an overall road-condition summary

Sample Output:

Potholes Detected: 4
Severe: 1
Moderate: 2
Minor: 1
Overall Road Condition: Poor

This is slightly more substantial than simple image classification because you have to combine **object detection + result analysis**.

3. Plant Disease Detection

Problem: Identify diseases in plants from leaf images to assist farmers or gardeners in early disease detection.

Streamlit:

- Upload plant/leaf image
- Run disease detection
- Display the uploaded image
- Show predicted disease
- Display confidence score
- Show healthy/diseased status
- Provide basic information about the detected disease
- Display recommended next steps or treatment category

4. Smart Waste Classification

Problem: Assist users in identifying the appropriate waste category.

The Streamlit interface could allow users to upload an image and receive:

Predicted Category: Plastic
Confidence Score: 94%
Suggested Disposal: Recyclable Waste

Note: If you opt classification problem, Make sure the problem should be genuine,

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